

corresponding to the nonminimal dependence (“the segment $[2, 4]$ intersects the triangle $[1, 3, 5]$ ”) and to the two minimal dependences (“the segment $[2, 4]$ intersects the segment $[1, 3]$, respectively the segment $[3, 5]$ ”) that we have calculated above.

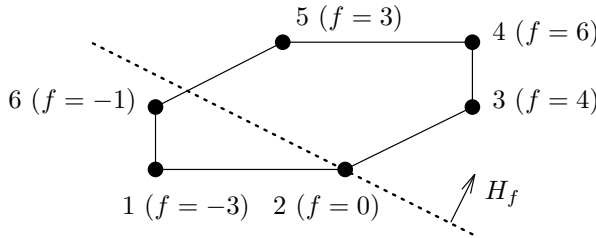
The sets $\mathcal{C}(X)$ and $\mathcal{V}(X)$ of signed circuits and vectors thus are combinatorial data associated with any affine point configuration. We will now proceed to extract the natural “dual data,” called signed cocircuits and signed covectors. In Section 6.2 we will derive the theoretical framework that shows that these four types of data are all “equivalent,” and that makes the “duality” precise.

(b) Affine Functions

From every affine function f on $\mathbb{R}^d$, of the form $\mathbf{x} \mapsto f(\mathbf{x}) = \mathbf{c}\mathbf{x} - z$, for $\mathbf{c} \in (\mathbb{R}^d)^*$, $z \in \mathbb{R}$, we get a row vector, called an *affine value vector*,

$$(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)) = (\mathbf{c}\mathbf{x}_1 - z, \dots, \mathbf{c}\mathbf{x}_n - z) = \mathbf{c}X - z\mathbb{1},$$

which records the values of $\mathbf{c}\mathbf{x} - z$ on the points $\mathbf{x}_i$ of X . For the hexagon from our example, the affine function $f(\mathbf{x}) = x_1 + 2x_2 - 3$ generates the affine value vector $(-3, 0, 4, 6, 3, -1)$:



where our drawing lists the values of f in brackets, and illustrates the hyperplane $H_f := \{\mathbf{x} \in \mathbb{R}^d : f(\mathbf{x}) = 0\}$ as a dashed line.

We note that the set

$$\text{a-Val}(X) := \{\mathbf{c}X - z\mathbb{1} \in (\mathbb{R}^n)^* : \mathbf{c} \in (\mathbb{R}^d)^*, z \in \mathbb{R}\}$$

of affine value vectors is a vector subspace of $(\mathbb{R}^n)^*$.

Geometrically, the row vector $f(X) := \mathbf{c}X - z\mathbb{1}$ records “signed distances” of the points $\mathbf{x}_i$ from the oriented hyperplane H_f . Thus the row sign vector $\text{sign}(f(X))$ records which points of X are on the positive side $H_f^+ \setminus H_f = \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) > 0\}$ of the hyperplane H_f , on H_f itself, or on the negative side of H_f .

For example, for the hexagon and the affine function illustrated above, we have $f(X) = (-3, 0, 4, 6, 3, -1)$, and thus $\text{sign}(f(X)) = (-, 0, +, +, +, -)$.

Definition 6.2. Let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathbb{R}^d$ be a finite set of n points in affine space $\mathbb{R}^d$.

The *signed covectors* of X are the row sign vectors $\mathbf{c}X - z\mathbf{1}$ corresponding to affine functions of the points in X ,

$$\mathcal{V}^*(X) := \{\text{sign}(\mathbf{c}X - z\mathbf{1}) : \mathbf{c} \in (\mathbb{R}^d)^*, z \in \mathbb{R}\} = \text{SIGN}(\text{a-Val}(X)).$$

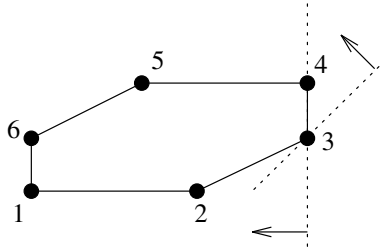
The *signed cocircuits* of X are the signed covectors of minimal support, for which the hyperplane $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{c}\mathbf{x} - d = 0\}$ is spanned by points in X . The set of signed cocircuits of X is denoted by $\mathcal{C}^* = \mathcal{C}^*(X)$.

In particular, it is easy to read off the faces of the polytope $\text{conv}(X)$ from the signed covectors. For this we identify each face of P with the set of vertices *not* on it, that is, the face F is associated with the *coface* $\text{vert}(P) \setminus \text{vert}(F)$. Similarly, we define a *cofacet* as the set of all vertices *not* on one facet. Thus the cofacets are the minimal (under inclusion) nonempty cofaces, and the cofaces are exactly all the unions of cofacets.

By slight abuse of language we will call a sign vector *positive* if it is nonnegative and nonzero, that is, if it lies in $\{0, +\}^n \setminus \mathbf{0}$, and similarly for row sign vectors. So we talk about “positive signed covectors,” which correspond to the nonvanishing nonnegative affine functions on a point configuration.

So, if $X \subseteq \mathbb{R}^d$, then the cofaces of $\text{conv}(X)$ are the supports of the positive covectors in $\mathcal{V}^*(X)$. Moreover, the cofacets of $\text{conv}(X)$ are the supports of the positive cocircuits in $\mathcal{C}^*(X)$. In particular, the face lattice of $\text{conv}(X)$ can be read off from $\mathcal{V}^*(X)$ as the set of all supports of positive covectors, ordered by inclusion. It can similarly be determined from $\mathcal{C}^*(X)$, since the cofaces are exactly the unions of cofacets.

For example, the drawing



illustrates affine functions $-x_1 + 5$ and $-x_1 + x_2 + 4$ that determine the positive covectors

$$(+, +, 0, 0, +, +) \quad \text{and} \quad (+, +, 0, +, +, +),$$

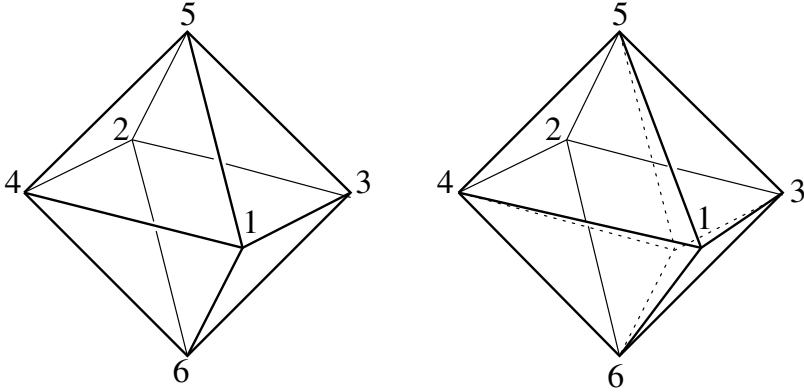
corresponding to the cofaces $\{1, 2, 5, 6\}$ and $\{1, 2, 4, 5, 6\}$, and thus to the vertex $\{3\}$ and to the edge $\text{conv}\{3, 4\}$. In principle, one can also read this off from the circuits (they are determined by the cocircuits, see next section), but this is not so straightforward.

Example 6.3 (Two octahedra).

Let $P_1 = C_3^\Delta = \text{conv}\{e_1, -e_1, e_2, -e_2, e_3, -e_3\}$ be the regular octahedron in $\mathbb{R}^3$, and let P_2 be obtained by perturbing the vertex e_1 to $e_1 + \frac{1}{6}e_2$. P_2 is a nonregular octahedron, that is, P_1 and P_2 are combinatorially equivalent, but not affinely isomorphic.

In matrices, we get $P_1 = \text{conv}(X_1)$ and $P_2 = \text{conv}(X_2)$, for

$$X_1 = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix} \quad X_2 = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ \frac{1}{6} & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$



We are not patient enough to list all the vectors and covectors. Here, however, are complete lists of all the cocircuits: the reader is not expected to check their details, but just to convince him/herself how this is constructed “from the picture,” and that it “seems correct.” Does it?

$$\begin{array}{ll} \pm(0 \ 0 \ 0 \ 0 \ + \ -), & \pm(0 \ 0 \ 0 \ 0 \ + \ -), \\ \pm(0 \ 0 \ + \ - \ 0 \ 0), & \pm(+ \ 0 \ + \ - \ 0 \ 0), \\ \pm(0 \ + \ 0 \ + \ 0 \ +), & \pm(0 \ + \ + \ - \ 0 \ 0), \\ \pm(0 \ + \ 0 \ + \ + \ 0), & \pm(0 \ 0 \ + \ - \ - \ 0), \\ \pm(0 \ + \ + \ 0 \ 0 \ +), & \pm(0 \ 0 \ + \ - \ 0 \ -), \\ \mathcal{C}^*(X_1): \pm(0 \ + \ + \ 0 \ + \ 0), & \pm(0 \ + \ 0 \ + \ 0 \ +), \\ \pm(+ \ 0 \ 0 \ + \ 0 \ +), & \pm(0 \ + \ 0 \ + \ + \ 0), \\ \pm(+ \ 0 \ 0 \ + \ + \ 0), & \pm(0 \ + \ + \ 0 \ 0 \ +), \\ \pm(+ \ 0 \ + \ 0 \ 0 \ +), & \pm(0 \ + \ + \ 0 \ + \ 0), \\ \pm(+ \ 0 \ + \ 0 \ + \ 0), & \pm(+ \ 0 \ 0 \ + \ 0 \ +), \\ \pm(+ \ - \ 0 \ 0 \ 0 \ 0), & \pm(+ \ 0 \ 0 \ + \ + \ 0), \\ & \pm(+ \ 0 \ + \ 0 \ 0 \ +), \\ & \pm(+ \ 0 \ + \ 0 \ + \ 0), \\ & \pm(+ \ - \ 0 \ 0 \ 0 \ 0), \end{array}$$

Similarly, the circuits are the columns of the matrices we describe next, and their negatives:

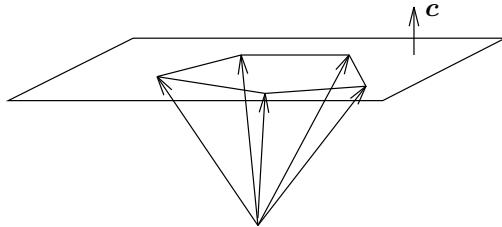
$$\mathcal{C}(X_1) : \begin{pmatrix} + & + & 0 \\ + & + & 0 \\ - & 0 & + \\ - & 0 & + \\ 0 & - & - \\ 0 & - & - \end{pmatrix} \quad \mathcal{C}(X_2) : \begin{pmatrix} + & + & + & 0 \\ + & + & + & 0 \\ - & - & 0 & + \\ - & 0 & + & + \\ 0 & - & - & - \\ 0 & - & - & - \end{pmatrix}.$$

6.2 Vector Configurations

While our discussion in Section 6.1 was on affine point configurations in $\mathbb{R}^d$, we now proceed to linear configurations of vectors in $\mathbb{R}^{d+1}$. The transition is the obvious one: with any configuration of points $\mathbf{x}_i$ in $\mathbb{R}^d$ (such as the vertices of a d -polytope), we associate the vectors $\mathbf{v}_i := \begin{pmatrix} 1 \\ \mathbf{x}_i \end{pmatrix}$ in $\mathbb{R}^{d+1}$. To get our notation for dimensions straight, we introduce a new parameter, called *rank*, as $r := d + 1$. As so often when dealing with a transition from affine to linear, it is convenient to have an extra letter r for affine rank (i.e., linear dimension), which is one more than the affine dimension d . Thus we have vectors $\mathbf{v}_i \in \mathbb{R}^r$.

In fact, what we get this way is an *acyclic vector configuration* $V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subseteq \mathbb{R}^r$, characterized by the following two properties (which are equivalent by a simple application of Farkas lemma II):

- (i) There is no nonnegative dependence, i.e., no $\mathbf{y} \geq \mathbf{0}$, $\mathbf{y} \neq \mathbf{0}$, such that $V\mathbf{y} = \mathbf{0}$.
- (ii) There is a linear function $\mathbf{c} \in (\mathbb{R}^r)^*$ such that $\mathbf{c}V > \mathbf{0}$ (i.e., $\mathbf{c}\mathbf{v}_i > 0$ for all i).



The construction of the sets of signed circuits, vectors, cocircuits, and covectors works for general vector configurations, acyclic or not. In fact, the linear algebra becomes slightly simpler in the linear case.

For this, let $V \in \mathbb{R}^{r \times n}$ be the matrix of a set of n vectors in $\mathbb{R}^r$. There may be multiple vectors, but we assume that the vectors span $\mathbb{R}^r$, so that $\text{rank}(V) = r$.